

[illegible][illegible]

3V3 SW10 SW_Push R62 10k 2.2n cap placed near ESP32 PWR_BUTTON R76 100k GND

The diagram shows five mounting holes, labeled H1 through H5, each with a red circle and a downward-pointing arrow indicating a connection to ground (GND). H1, H2, H3, and H4 are connected directly to GND. H5 is connected to GND through a 10k resistor, represented by a red circle with a diagonal line through it.

[illegible]

High resistor values -> low idle current

ratio will be 0.5
4.2V max -> 2.1V
3.3V min -> 1.65V

BAT_MONIT

1u cap placed at ESP32

R12
1M

R10
1M

PWR_FLAG

GND

[illegible]

Pinout diagram for J6 connector of the PPPC062LJ8N-RC module:

Pin	Signal
1	GND
2	UNUSUED_GPIO_46
3	UNUSUED_GPIO_45
4	GND
5	LED_SW
6	I2C_SDA
7	I2C_SCL
8	UNUSUED_GPIO_3
9	UNUSUED_GPIO_4
10	UNUSUED_GPIO_5
11	C
12	P+

Module Label: J6 PPPC062LJ8N-RC

The schematic diagram illustrates the pin connections for the ESP32 module. It includes connections for unused GPIO pins (45, 3, 46, 4), I2C pins (SCL, SDA), and SD card pins (DAT1, DAT0). Key components connected include a TPDA4E1U06DBVR USB-to-UART bridge, a SMAJ26A diode, a LED_SW switch, a PESD2VNN-UX ESD protection diode, and a TSD05CDYFR temperature sensor. Power and ground connections are clearly marked, including a 3V3 supply and a 2.2nF capacitor near the ESP32.

Two circuit diagrams illustrating button connections to an ESP32 microcontroller. Both diagrams show a 3V3 power source and a 10k pull-up resistor connected to the button pins.

Left Diagram: Shows four buttons connected to pins R6D, R4B, R49, and R2D. A 2.2nF capacitor is placed near ESP32. The buttons are labeled SW2 (SW_Push), SW3 (SW_Push), SWB (SW_Push), and SW9 (SW_Push). The ground connections are labeled GND RIGHT, GND LEFT, GND CONFIRM, and GND BACK.

Right Diagram: Shows four buttons connected to pins R6B, R11, R15, and R73. A 2.2nF capacitor is placed near ESP32. The buttons are labeled SW1 (SW_Push), SW4 (SW_Push), SW5 (SW_Push), and SW7 (SW_Push). The ground connections are labeled GND DOWN(1), GND UP(1), GND DOWN(2), and GND UP(2). A note indicates that UP(2) can serve as a power button if R36/R73 are unpopulated and R72/R74 are populated.

ratio will be 0.5
3.3V min -> 1.65V

TP3 TestPoint LED_SW
TP4 TestPoint C-
TP5 TestPoint W-

J3
FH34SRJ-65-0.55H_50

LED_SW 1 C+
2 C-
3 NC
4 NC
5 W+
6 W-

SHIELD S1

Powered via LDO_IN source
Instead of 3V3 to reduce stress on LDO output power

GND

LDO_IN

PWM_LED

U10
TP5923610DRLR

1 VIN
2 ADIM
3 FB
4 SW
5 VOUT
6 GND

1u/50V
C9

LED_SW

GND

3V3

100n
C24

GND

COLOR_SEL

R75
100k

GND

U12
74LVC1604

5 VCC
3 GND
4 COLOR_SEL_INV

GND

Q6
B5S138

1 COLOR_SEL_INV
2 C-
3 W-

R49
1M

GND

Q5
B5S138

1 COLOR_SEL
2 W-
3 R50
4 1M

GND

TP5923 chip is a series LED boost converter.
Recommended to run between 10kHz-25kHz via PWM_LED for brightness control.
Return path for LEDs is toggled via R37 for warm/cool selection.
Feedback current is 15mA tuned via R37. Reference datasheet to change as needed.
LED_MONIT is analog signal that references 0-24.5V output on LED voltage. (ADC range from 0 to -30V)
Internal OVP should cut out at -24V, but software protection should kick in and disable LED_ACTIVATE at safe voltage above known forward voltage (15V)

The diagram shows two push-button switches, SW6 and SW11, connected to the ESP32_I00 and ESP32_EN pins respectively. Each switch is connected to a 100 ohm resistor (R64 and R63) and has its other terminal connected to GND.

[illegible]

Typical Configuration used on 4.26"

Alternate Configuration (rarely used, do not populate by default)